



Setup Guide

Local installation, credentials, and running the agent

Everything needed to run Career Trajectory Optimizer end to end: environment setup, provider credentials, the notebook walkthrough, and the Streamlit UI.

Python 3.12+

uv

OpenAI required

LinkUp/Tavily optional

~10 min

PREREQUISITES



Python 3.12+

A working interpreter; the project ships a pyproject.toml managed by uv.



uv

Handles the virtual environment and locked dependency installs via uv sync.



OpenAI API key

Required — the orchestrator and every subagent call this model.



LinkUp or Tavily key (optional)

Enables market-researcher's live web search; omit both to run grounding-only.

1. CLONE & INSTALL

```
# 1. Enter the project
cd careertrajectoryoptimizer

# 2. Install dependencies (creates .venv)
uv sync

# 3. Create your local environment file
cp .env.example .env
```

2. CONFIGURE CREDENTIALS

Copy `.env.example` to `.env` and fill in the values below. Only `OPENAI_API_KEY` is required.

VARIABLE	PURPOSE	EXAMPLE
OPENAI_API_KEY	Required — powers the orchestrator and all subagents	sk-proj-...
LINKUP_API_KEY	Primary web-search provider for market-researcher	your-linkup-key
TAVILY_API_KEY	Fallback web-search provider if LinkUp is not set	your-tavily-key
CAREER_AGENT_MODEL	Overrides the default model	gpt-5-mini
LANGSMITH_API_KEY	Optional nested trace tree tracing	lsv2_pt_...

3. RUN THE APP

```
# From the project root
uv run streamlit run app.py

# Open in a browser
http://localhost:8501
```

4. OR RUN THE NOTEBOOK

```
# In VS Code / Jupyter, select the project's .venv as the kernel
career_optimizer_deep_agent.ipynb

# Run the cells top to bottom
```

5. SUBAGENTS AT A GLANCE

SUBAGENT	ROLE	TOOLS
profile-analyzer	Parses profile/resume into a structured timeline	filesystem only
market-researcher	Gathers live Indian tech market data	web_search (LinkUp/Tavily)
salary-intelligence	Benchmarks pay vs. market	reads research brief
market-tracker	Tracks skill demand & trends	reads research brief
risk-scorer	Scores 5-axis risk dashboard	risk_dashboard_validator
career-recommender	Goal-based gap analysis (Aspiration Mode)	reads all analysis
roadmap-planner	Builds the 30-60-90 day roadmap	roadmap_validator
critic	Reviews grounding, methodology, consistency	both validators

TROUBLESHOOTING



OPENAI_API_KEY missing

The Streamlit sidebar shows a red status and blocks the agent from building; add it to `.env` and refresh.



A single call stalls for minutes

Likely a per-minute rate limit; the model is bounded to `max_retries=2, timeout=60s` so it fails fast and visibly instead.



A run takes 5+ minutes

Expected — 6 to 9 subagents each run their own multi-turn loop. Career Aspiration Mode and the calendar-sync HITL gate add more time.



Resume text looks incomplete

Many resumes lay out skills/experience in tables — the extractor walks the whole document (paragraphs + nested tables) to avoid missing them.

Ready to run in about 10 minutes.

No infrastructure to stand up — the store, checkpointer, and backend are all in-process; only your API keys are external.

- 1. uv sync
- 2. Configure .env
- 3. streamlit run
- 4. Upload a resume